

Sales Pulse

Sales and Demand Forecasting System

REVIEW – 2

Batch 1 - Section 5

Team members

Roll No	Name
2420030427	Y. Nimisha
2420030498	K. Sanuhya
2420030516	R. Sarvajith varma

Under the guidance of

Dr. N Sunanda

Associate Professor

Department of Computer Science and Engineering



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Table of contents

- Problem statement
- Objectives
- Literature Survey/ Existing systems

Problem statement

- Businesses need to analyze historical sales data to understand sales trends and demand variations.
- Traditional sales management and reporting systems mainly record transactions and display past sales information.
- These systems have limited ability to predict future sales and demand.
- Businesses may face difficulties in identifying high- and low-demand periods and planning inventory effectively.
- Sales Pulse aims to solve this problem by using data science and time-series forecasting to predict future sales and support better business decisions.

Real-World Relevance:

- Helps businesses plan inventory and sales.
- Identifies seasonal patterns and demand variations.
- Supports informed and proactive decision-making.

Motivation:

To develop a data science-based system that transforms historical sales data into useful future sales predictions and business insights.

Objectives

- To collect and analyze historical sales data.
- To preprocess data by handling missing, inconsistent, and incorrect records.
- To identify sales trends, seasonal patterns, and demand variations through exploratory data analysis.
- To apply time-series forecasting models such as ARIMA or Prophet to predict future sales and demand.
- To evaluate forecasting performance using suitable accuracy metrics.
- To develop a user-friendly dashboard to display historical sales, forecasted values, and important insights.
- To support better inventory planning and informed business decision-making.

Expected Outcomes

- Accurate and useful future sales predictions.
- Clear visualization of sales trends and forecasted demand.
- Identification of high- and low-demand periods.
- Better understanding of sales patterns.
- Improved support for inventory and sales planning.

Literature Survey/ Existing System

S.No	Research Paper	Author(s)	Link	Drawback
1.	Time-series forecasting of seasonal items sales using machine learning – A comparative analysis	Research paper authors	https://doi.org/10.1016/j.jji.mei.2022.100058	Focuses on furniture sales data. The results may not generalize to all product categories and businesses.
2.	A Comparative Study on Forecasting of Retail Sales	Md Rashidul Hasan, Muntasir A Kabir, Rezoan A Shuvro, Pankaz Das	https://arxiv.org/abs/2203.06848	Uses Walmart historical sales data. The performance may vary for other businesses and datasets.
3.	Predictive Analytics for Demand Forecasting – A Comparison of SARIMA and LSTM in Retail SCM	Research paper authors	https://doi.org/10.1016/j.procs.2022.01.298	Uses sales data from a specific Austrian retailer. Model performance depends on product demand patterns and promotions.
4.	Sales Forecasting Models: Comparison between ARIMA, LSTM and Prophet	Dmitry Brykin	https://doi.org/10.3844/jcssp.2024.1222.1230	Evaluates specific sales categories from banking data. The best model may differ for other businesses and forecasting periods.
5.	Research on Precise Demand Forecast of Retail Commodities Based on LSTM-ARIMA Joint Model	Haoran Song, Dengyu Du, Yao Li	https://doi.org/10.1088/1742-6596/1865/4/042049	Focuses on a joint LSTM-ARIMA model. Combining models can increase complexity and computational requirements.
6.	Short-Term Sales Forecasting Using LSTM and Prophet Based Models in E-Commerce	Alp Ecevit, İrem Öztürk, Mustafa Dağ, Tuncay Özcan	https://dergipark.org.tr/en/pub/acin/article/1259067	Focuses on short-term e-commerce sales forecasting. The results may not apply to long-term forecasting or other industries.